Medicare Payment Gap Analysis in Massachusetts

A Multi-Method Analysis of 37,708 Providers Using 2023 CMS Data

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Executive Summary

Every time a Medicare provider sees a patient covered under Part B, they submit a charge to the Centers for Medicare and Medicaid Services (CMS), which pays a fraction of that amount. The difference, the payment gap, varies enormously across providers. In Massachusetts alone, the median gap is approximately $89,292 per provider annually, with some providers exceeding $100 million. This project uses the 2023 CMS Medicare Physician & Other Practitioners dataset, filtered to 37,708 Massachusetts providers, to identify what drives this variation using four statistical methods: multiple linear regression (MLR), one-way ANOVA, logistic regression, and bootstrap and permutation tests. Each analysis is accompanied by robustness checks that stress-test the findings; all core conclusions hold or strengthen under scrutiny.

Provider volume is the dominant driver of the gap. Each doubling of beneficiaries served is associated with a 1.96-fold larger gap (MLR, Model 1, $R^2 = 0.63$). Provider volume effects are uniform across the gap distribution, while billing intensity and patient complexity matter more for low-gap providers, a heterogeneity OLS averages over.

Specialty group membership independently explains 21% of variance in log payment gap (Welch's ANOVA, $F(9, 6{,}181.8) = 929.99$, $\eta^2 = 0.21$). Diagnostic Radiology, Anesthesiology, and Emergency Medicine anchor the top among the ten largest specialties, while Licensed Clinical Social Workers and Psychiatry anchor the bottom. A descriptive analysis of all 97 provider specialties reveals that Diagnostic Radiology ranks approximately 20th in the full landscape. Facility and equipment-based entities (non-physician organizational billers) occupy a separate tier with gaps one to two orders of magnitude larger, driven by per-procedure unit costs rather than billing breadth (distinct services billed). At the lower end, specialties with narrow standardized billing profiles, vaccine administrators and allied health providers, consistently bill at near-Medicare rates.

Provider volume is also the single strongest predictor of high-gap classification (OR = 6.72, $z = 66.34$), followed by practice breadth (OR = 3.77) and billing intensity (OR = 2.65).

Providers serving higher proportions of dual-eligible patients, those covered by both Medicare and Medicaid, are substantially less likely to be high-gap classifiers (OR = 0.21). This dual-eligibility finding is the most robust result in the paper and holds across threshold definitions, specialty compositions, and continuous specifications. The logistic model achieves AUC = 0.905 and 82.5% accuracy using six readily observable variables. Providers serving low dual-eligibility panels have a median gap $76,628 higher than those serving high dual-eligibility panels ($p < 0.0001$, permutation test).

The findings suggest that the payment gap is structured, predictable, and robust to analytical choices. A six-variable screening model can correctly classify four in five Massachusetts providers without claims-level audit data, offering a practical triage tool for Medicare billing oversight. Oversight efforts should prioritize procedurally intensive specialties and high-volume providers. The consistent association between dual-eligible patient panels and smaller gaps, which holds across threshold definitions, specialty compositions, and continuous specifications, warrants direct policy investigation. The payment gap is not a mystery: it has predictors, a specialty profile, and a patient-panel signature that holds up under scrutiny.

1 Introduction

Despite spending more on health care than any other wealthy nation, over 16% of GDP as of 2022, a figure projected to exceed 20% by 2035, the United States consistently ranks last among high-income countries on measures of access, equity, and health outcomes (Blumenthal et al., 2024). A central driver of this paradox is not how much care Americans receive, but how unevenly resources are distributed across payers and providers. Private insurers, for instance, pay substantially more than Medicare for the same services (Lopez et al., 2020), creating a fragmented reimbursement landscape in which provider payment varies widely depending on who is footing the bill. More recent work published in *JAMA Network Open* found that social determinants of health account for nearly 37.7% of the variation in price-adjusted Medicare spending across U.S. counties (Zhang et al., 2021). These disparities are not merely

geographic: they reflect differences in how providers bill, who their patients are, and how practice patterns interact with Medicare's payment structure.

This project investigates those disparities at the provider level using the 2023 CMS Medicare Physician & Other Practitioners public use dataset, filtered to Massachusetts providers (n= 37,708). The primary response variable is the *payment gap*: the difference between total submitted charges (`Tot_Sbmtd_Chrg`) and CMS standardized payments (`Tot_Mdcr_Stdzd_Amt`). While submitted charges are known to exceed actual costs, systematic variation in the gap across providers reflects meaningful differences in billing behavior and practice patterns (Lopez et al., 2020). Standardized payments remove geographic adjustments for local wages and input prices, ensuring that observed variation reflects provider practice patterns rather than CMS pricing artifacts (O'Donnell et al., 2013). Massachusetts, with an uninsured rate of just 2.1%, compared to 8.2% nationally (Center for Health Information and Analysis, 2023), offers an instructive case study: if payment disparities persist here despite near-universal coverage, provider-level practice patterns are likely a primary driver rather than structural gaps in insurance access. Massachusetts also hosts a high concentration of academic medical centres and procedural subspecialists, making it a particularly informative case for examining how specialty mix and billing intensity interact with Medicare's payment structure.

Four research questions are addressed, each with a distinct statistical method.

RQ1 (Multiple Linear Regression): Do provider volume, patient risk profile, and chronic disease burden predict the standardized Medicare payment gap among Massachusetts providers, and which factors have the strongest independent effect holding all others constant? Quantile regression extensions at the 25th, 50th, and 75th percentiles complement both OLS models to assess whether predictor effects are uniform across the payment gap distribution.

RQ2 (One-Way ANOVA): Do average standardized Medicare payment gaps differ significantly across provider specialties in Massachusetts, and which specific specialties drive the largest gaps? Welch's one-way ANOVA is applied across the top 10 specialties by provider

count, followed by Games–Howell post-hoc tests and a non-parametric robustness check using Kruskal–Wallis and Dunn's test. A descriptive analysis of all 97 provider specialties contextualises the formal ANOVA findings within the full Massachusetts specialty landscape.

RQ3 (Logistic Regression): Do provider structure, billing behavior, and dual-eligibility beneficiary proportion predict whether a Massachusetts provider falls in the high payment gap group? Logistic regression predicts a binary outcome `high_gap`, defined as a standardized payment gap above the Massachusetts median ($89,292).

RQ4 (Bootstrap & Permutation): Does the median standardized payment gap differ significantly between providers whose patient panels have low versus high proportions of dual-eligible beneficiaries, and what is a robust confidence interval for that difference? Providers are split at the median `pct_dual` into two equal groups, and the unconditional gap between them is assessed without covariate adjustment.

2 Data & Methods

2.1 Dataset

The data come from the 2023 CMS Medicare Physician & Other Practitioners by Provider dataset, publicly available at https://data.cms.gov/provider-summary-by-type-of-service/medicare-physician-other-practitioners (Centers for Medicare & Medicaid Services, 2023). The full national dataset was filtered to Massachusetts providers, yielding 37,713 observations across 81 variables. Each row is a provider-level annual summary record identified by National Provider Identifier (NPI), covering all Medicare Part B fee-for-service activity submitted during 2023. The dataset excludes Medicare Advantage (Part C) enrollees and suppresses records derived from 10 or fewer beneficiaries.

Response variable. `payment_gap` = `Tot_Sbmtd_Chrg` − `Tot_Mdcr_Stdzd_Amt`, where standardized payments remove geographic wage and input price adjustments, making gaps comparable across Massachusetts geographies (O'Donnell et al., 2013). A log transformation

(log_payment_gap) is applied for regression to address severe right skew.

Key predictors. Tot_Benes (practice size; log-transformed in regression models to address right skew), srvcs_per_bene = Tot_Srvcs / Tot_Benes (billing intensity; log-transformed), Bene_Avg_Risk_Scre (HCC risk score; higher = sicker panel), Bene_Avg_Age, and 25 chronic condition prevalence variables (Bene_CC_*, range 0–75 after CMS top-coding at 75%).

Grouping variable. Rndrng_Prvdr_Type (provider specialty, 97 levels; formal ANOVA restricted to top 10 by provider count; all 97 levels included in a descriptive ranked analysis).

Binary outcome. high_gap = 1 if payment_gap exceeds the Massachusetts median ($89,292), else 0.

RQ3/RQ4 predictors. pct_dual = Bene_Dual_Cnt / Tot_Benes (proportion of dual-eligible, i.e. low-income, beneficiaries; used as a binary split at the median for RQ4 and examined continuously to validate that split); drug_provider = 1 if the provider submitted any drug-related charges, else 0; Tot_HCPCS_Cds (procedure diversity).

2.2 Data Quality & Preprocessing

The payment gap is heavily right-skewed (median $89,292; mean $247,753; max > $188M). Five providers with negative payment gaps were removed, yielding a final dataset of 37,708 providers. Bene_CC_* values are suppressed by CMS when fewer than 11 beneficiaries with a given condition are observed, leaving complete chronic condition data for only 3,752 providers (9.9%). Suppression is concentrated among low-volume providers. This motivates a two-model design for RQ1: **Model 1** uses the full 37,708-provider dataset with Bene_Avg_Risk_Scre as the sole complexity control; **Model 2** uses the 3,752-provider complete-CC subset with LASSO-selected chronic condition variables added.

Bene_Dual_Cnt is suppressed for 9,152 providers (24.3%); complete-case analysis retains 28,556 providers for RQ3 and RQ4. Because CMS suppresses Bene_Dual_Cnt when a provider has 10 or fewer dual-eligible beneficiaries, excluded providers are systematically small-panel

practitioners; RQ3 and RQ4 findings therefore generalise to providers with at least moderate patient volumes (see Section 5).

Missing `Drug_Sbmtd_Chrg` (3,403 providers, 9.0%) indicates no drug billing and is zero-imputed before deriving `drug_provider`. Geographic analysis was ruled out after RUCA classification revealed that 98.0% of providers are in metropolitan core ZIP codes. `set.seed(6105)` was applied to all bootstrap and permutation procedures.

2.3 Statistical Methods

Multiple Linear Regression (RQ1). MLR estimates the independent contribution of each predictor to `log_payment_gap` while holding others constant. This is essential for disentangling volume, billing intensity, and patient complexity. VIF is computed to check multicollinearity; LINE assumptions are assessed via diagnostic plots. Quantile regression at the 25th, 50th, and 75th percentiles complements both OLS models to assess whether predictor effects are uniform across the conditional payment gap distribution, using kernel-based standard errors (`se = "ker"`).

One-Way ANOVA (RQ2). ANOVA tests whether mean `log_payment_gap` differs across specialty groups in a single omnibus F-test, controlling Type I error inflation that would arise from pairwise t-tests. Because Levene's test rejects equal variances, Welch's ANOVA is used with Games–Howell post-hoc tests. Effect size is reported as η^2. A non-parametric robustness check using Kruskal–Wallis with Dunn's post-hoc test (Bonferroni correction) confirms whether parametric and rank-based approaches agree on the specialty tier structure. A descriptive ranked analysis of all 97 provider specialties contextualizes the formal ANOVA findings within the full Massachusetts specialty landscape.

Logistic Regression (RQ3). The binary outcome `high_gap` requires logistic regression; OLS would produce probabilities outside $[0, 1]$ and violate constant-variance assumptions. Exponentiated coefficients yield odds ratios interpreted as the multiplicative change in high-gap odds per unit increase in each predictor, holding all others constant. Model performance

is assessed via confusion matrix, sensitivity, specificity, precision, and ROC/AUC. Two robustness checks are applied: a sensitivity analysis refitting the model at the 40th, 60th, and 75th percentile classification thresholds to assess whether findings depend on the median split definition, and a specialty-controlled refit adding specialty group dummies to assess whether the dual-eligibility finding reflects specialty composition rather than an independent association.

Bootstrap & Permutation (RQ4). The heavily skewed payment gap distribution makes parametric CIs for the median unreliable. Bootstrap resampling constructs an empirical CI for the difference in median payment gaps between low and high dual-eligibility provider groups without distributional assumptions. A permutation test then assesses whether the observed difference exceeds what chance alone would produce under the null hypothesis of no group difference. A continuous analysis of `pct_dual` using univariate quantile regression validates the median split by confirming that the relationship between dual-eligibility burden and the payment gap is linear and uniform across the full range of `pct_dual`.

Statistical vs. Practical Significance. With $n = 37{,}708$, even trivially small effects will achieve statistical significance. Practical significance is therefore assessed via effect size measures throughout: R^2 and adjusted R^2 for MLR, η^2 for ANOVA, odds ratios and McFadden pseudo-R^2 for logistic regression, and the bootstrap confidence interval for the median difference in RQ4. Statistically significant predictors are flagged as practically meaningful only when the associated effect size warrants that interpretation.

All analyses were conducted in R (R Core Team, 2024) using the following packages: `ggplot2` (Wickham, 2016), `dplyr` (Wickham et al., 2023a), `broom` (Robinson et al., 2023), `glmnet` (Friedman et al., 2023), `pROC` (Robin et al., 2023), `car` (Fox and Weisberg, 2019), `boot` (Canty and Ripley, 2023), `rstatix` (Kassambara, 2023), `multcomp` (Hothorn et al., 2023), `multcompView` (Graves et al., 2023), `haven` (Wickham et al., 2023b), `corrplot` (Wei and Simko, 2023), and `quantreg` (Koenker, 2024).

3 Results

3.1 Multiple Linear Regression

Research Question: Do provider volume, patient risk profile, and chronic disease burden predict the standardized Medicare payment gap among Massachusetts providers, and which factors have the strongest independent effect holding all others constant?

Because CMS suppresses `Bene_CC_*` values for low-volume providers, complete chronic condition data is available for only 3,752 providers (9.9%). These providers differ systematically from the full population (median gap $404,539 vs. $89,292; median risk score 1.727 vs. 1.326), making zero-imputation and pooling untenable. Two complementary models are therefore estimated: **Model 1** uses the full dataset ($n = 37{,}708$) with volume, billing intensity, age, and risk score as predictors; **Model 2** uses the complete-CC subset ($n = 3{,}752$) and adds LASSO-selected chronic condition variables.

Model 1: Full Dataset ($n = 37{,}708$)

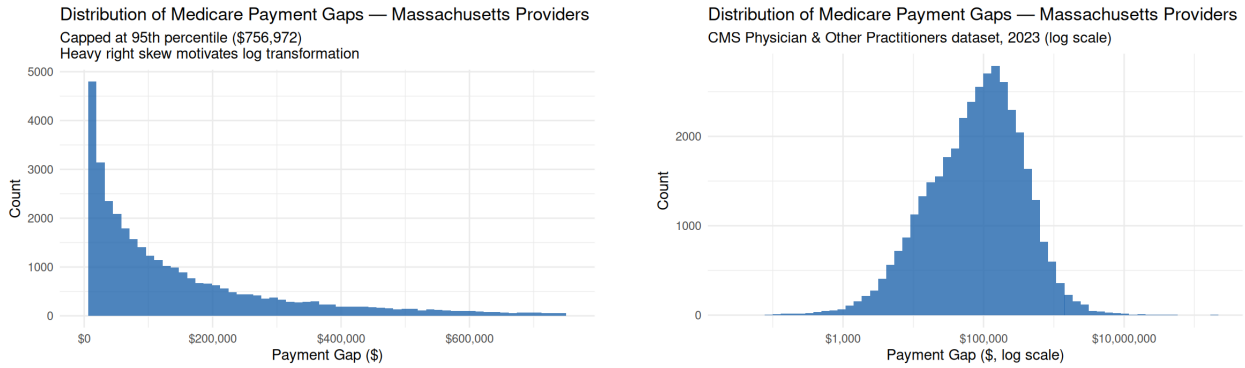

Figure 1: Raw (left, capped at 95th percentile) and log-transformed (right) payment gap distributions. Log transformation is applied to address severe right skew.

The estimated regression equation is:

$$\log(\widehat{\texttt{payment_gap}}) = 5.4257 + 0.9709 \cdot \log(\texttt{Tot_Benes}) + 0.4166 \cdot \log(\texttt{srvcs_per_bene})$$
$$+ 0.0011 \cdot \texttt{Bene_Avg_Age} + 0.3121 \cdot \texttt{Bene_Avg_Risk_Scre}$$

Three of four predictors are significant at $\alpha = 0.001$ ($p < 2 \times 10^{-16}$); `Bene_Avg_Age` is not significant in Model 1 ($p = 0.213$) but is retained as a standard demographic control; it reaches significance in Model 2 ($p < 0.001$) once chronic condition controls are added, suggesting its effect is absorbed by correlated predictors in the full dataset rather than absent.

Provider volume (`log(Tot_Benes)`) is the strongest predictor by t-value ($t = 226.5$): each doubling of beneficiaries served is associated with a 1.96-fold larger payment gap ($e^{0.9709} = 2.640$, 95% CI: [2.618, 2.663]), holding all else constant. Each doubling of services per beneficiary is associated with a $2^{0.417} \approx 1.34$-fold larger gap ($e^{0.4166} = 1.517$, 95% CI: [1.501, 1.532]). Patient complexity (`Bene_Avg_Risk_Scre`) contributes a 36.7% larger gap per one-unit increase in average HCC risk score ($e^{0.3121} = 1.366$, 95% CI: [1.348, 1.385]).

The model explains 62.9% of variance ($R^2 = 0.6291$, adjusted $R^2 = 0.6291$; $F(4, 37{,}703) = 15{,}989$, $p < 2.2 \times 10^{-16}$). All VIFs are below 1.22 (range: 1.04–1.21), confirming no multicollinearity.

Diagnostic Plots, Model 1

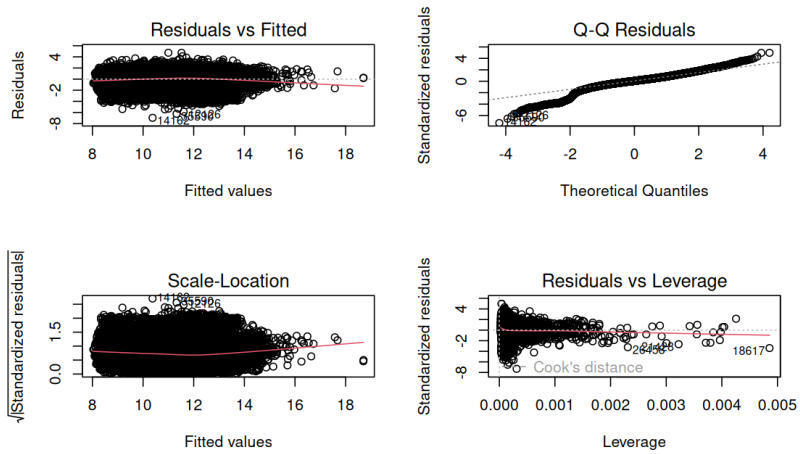

Figure 2: Model 1 diagnostic plots: Residuals vs. Fitted (top left), Normal Q-Q (top right), Scale-Location (bottom left), Residuals vs. Leverage (bottom right).

Three of the four LINE assumptions are violated. **Linearity** is substantially improved relative to the untransformed model but not fully resolved: the Residuals vs. Fitted smoother shows mild curvature at high fitted values. **Normality** fails: the Q-Q plot shows heavy left-tailed departures, with the most extreme residual reaching approximately -7 standardized units. **Equal variance** fails: the Scale-Location plot confirms residual spread decreases at higher fitted values. **Independence** is defensible: providers are separate billing entities in a cross-sectional dataset. Cook's distance flags 1,899 influential observations (5.0%). A robustness check confirmed that the core coefficients are stable after removing these observations, with `log(Tot_Benes)` and `log(srvcs_per_bene)` changing by less than 3%; `Bene_Avg_Risk_Scre` attenuates modestly from 0.312 to 0.256, consistent with the quantile regression finding that patient complexity matters less at higher quantiles where extreme providers are more prevalent.

Inference remains defensible: at $n = 37{,}708$, the CLT ensures asymptotically valid coefficient estimates and standard errors despite non-normality, and heteroscedasticity affects efficiency but not consistency of OLS estimates.

Quantile Regression Extension, Model 1

To assess whether the OLS coefficients mask heterogeneous effects across the payment gap distribution, Model 1 is complemented with a multivariate quantile regression at the 25th, 50th, and 75th percentiles, using the same four predictors and the full dataset ($n = 37{,}708$). Quantile regression coefficients are estimated at the 25th, 50th, and 75th percentiles of the *conditional* payment gap distribution, that is, the gap distribution after accounting for each provider's volume, billing intensity, age, and risk score. Q25 therefore represents under-billers relative to their profile, Q50 represents typical billers, and Q75 represents over-billers; the coefficients reveal whether the factors that drive over-billing are the same as those that drive under-billing, something OLS cannot determine.

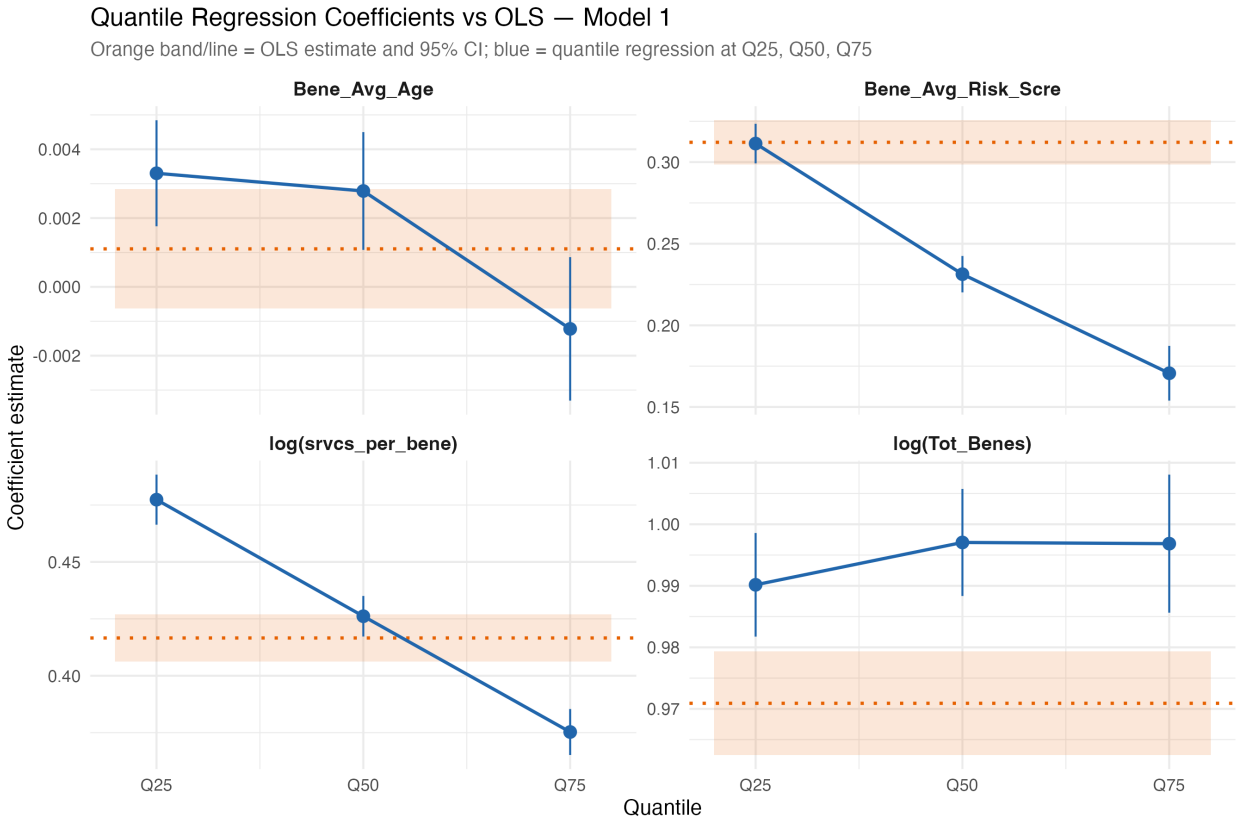

Figure 3: Multivariate quantile regression coefficients at Q25, Q50, and Q75 (blue) versus OLS estimate and 95% confidence interval (orange). Each panel shows one predictor; the x-axis indexes the quantile of the conditional payment gap distribution.

Table 1 reports the full coefficient estimates and standard errors at each quantile.

Table 1: Quantile regression coefficients for Model 1 at Q25, Q50, and Q75. Standard errors estimated via kernel method. All coefficients significant at $p < 0.001$ except `Bene_Avg_Age` at Q75 ($p = 0.251$).

	Q25		Q50		Q75	
Predictor	Estimate	SE	Estimate	SE	Estimate	SE
`log(Tot_Benes)`	0.990	0.004	0.997	0.004	0.997	0.006
`log(srvcs_per_bene)`	0.477	0.006	0.426	0.005	0.375	0.005
`Bene_Avg_Age`	0.003	0.001	0.003	0.001	−0.001	0.001
`Bene_Avg_Risk_Scre`	0.311	0.006	0.231	0.006	0.171	0.009

Quantile regression reveals two patterns OLS cannot detect. Provider volume (`log(Tot_Benes)`) is stable across quantiles (0.990, 0.997, 0.997), confirming that volume affects all providers similarly regardless of where they sit in the gap distribution. By contrast, billing intensity (`log(srvcs_per_bene)`) and patient complexity (`Bene_Avg_Risk_Scre`) decline monotonically from Q25 to Q75 ($0.477 \rightarrow 0.375$ and $0.311 \rightarrow 0.171$ respectively), both straying outside the OLS confidence band. OLS overstates the role of these predictors by averaging across this heterogeneity: among high-gap providers, billing intensity and patient complexity contribute relatively little once volume dominates. `Bene_Avg_Age` changes sign and is non-significant at Q75 ($p = 0.251$), confirming it is not a stable predictor across the distribution.

The practical implication for billing oversight follows directly: interventions targeting billing intensity and patient mix are likely more effective for mid-range providers than for the highest-gap billers, where volume is the primary structural driver.

Model 2: Complete-CC Subset ($n = 3{,}752$)

LASSO with 10-fold cross-validation (`lambda.1se`) retained 9 of 25 `Bene_CC_*` variables. After removing `Bene_CC_BH_Mood_V2_Pct` due to collinearity ($r = 0.87$ with arthritis; $r = 0.79$ with

dementia), 8 chronic condition predictors enter Model 2. The non-zero LASSO coefficients are shown in Figure 4.

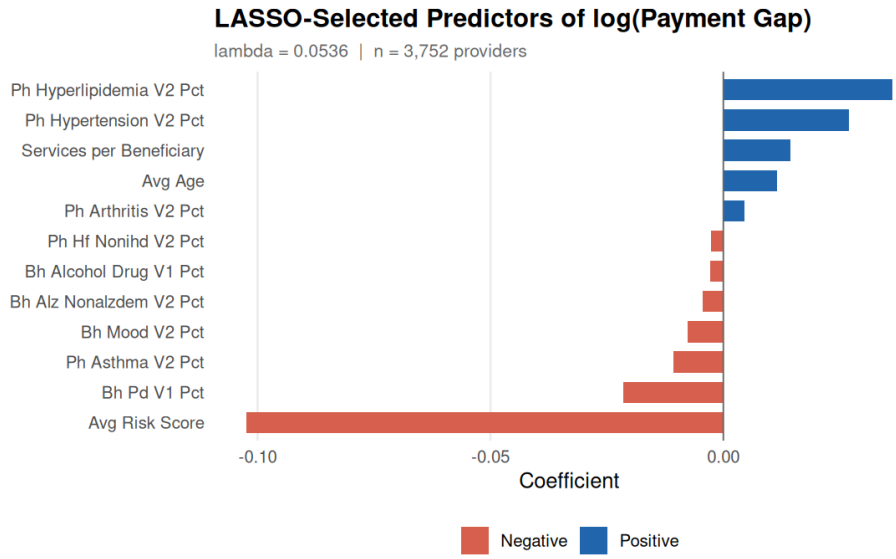

Figure 4: Non-zero LASSO coefficients at `lambda.1se`, ordered by absolute magnitude. These shrinkage coefficients are used for variable selection only; MLR coefficients differ in magnitude.

The estimated regression equation is:

$$
\begin{aligned}
\log(\widehat{\texttt{payment_gap}}) = & -0.6503 + 0.5552 \cdot \log(\texttt{Tot_Benes}) + 0.5236 \cdot \log(\texttt{srvcs_per_bene}) \\
& + 0.0419 \cdot \texttt{Bene_Avg_Age} + 0.1170 \cdot \texttt{Bene_Avg_Risk_Scre} \\
& + 0.0301 \cdot \texttt{Bene_CC_PH_Hyperlipidemia_V2_Pct} \\
& + 0.0523 \cdot \texttt{Bene_CC_PH_Hypertension_V2_Pct} \\
& + 0.0171 \cdot \texttt{Bene_CC_BH_PD_V1_Pct} \\
& - 0.0036 \cdot \texttt{Bene_CC_PH_Asthma_V2_Pct} \\
& + 0.0069 \cdot \texttt{Bene_CC_PH_Arthritis_V2_Pct} \\
& - 0.0178 \cdot \texttt{Bene_CC_BH_Alz_NonAlzdem_V2_Pct} \\
& + 0.0183 \cdot \texttt{Bene_CC_BH_Alcohol_Drug_V1_Pct} \\
& - 0.0117 \cdot \texttt{Bene_CC_PH_HF_NonIHD_V2_Pct}
\end{aligned}
$$

Ten of twelve predictors are significant at $\alpha = 0.05$. `Bene_Avg_Risk_Scre` ($p = 0.054$) is retained as a pre-specified complexity control carried forward from Model 1; its marginal significance reflects shared variance with the condition indicators (VIF = 6.05) rather than a genuine absence of effect. `Bene_CC_PH_Asthma_V2_Pct` ($p = 0.417$) was selected by LASSO and is retained on those grounds.

Key coefficient interpretations (holding all else constant): each doubling of beneficiaries served is associated with a 1.47-fold larger gap ($e^{0.5552} = 1.742$, 95% CI: [1.679, 1.808]); each doubling of services per beneficiary with a 1.44-fold larger gap ($e^{0.5236} = 1.688$, 95% CI: [1.638, 1.739]). Among chronic conditions, hypertension prevalence has the largest positive effect: each percentage point increase is associated with a 5.4% larger gap ($e^{0.0523} = 1.0537$, 95% CI: [1.044, 1.064], $p < 2 \times 10^{-16}$); hyperlipidemia adds 3.1% per point ($e^{0.0301} = 1.0306$, 95% CI: [1.022, 1.040]). Dementia (-1.8%, $p < 2 \times 10^{-16}$) and heart failure (-1.2%, $p < 0.001$) are negatively associated with the gap.

The model explains 48.5% of variance ($R^2 = 0.4855$, adjusted $R^2 = 0.4838$; $F(12, 3{,}739) = 294.0$, $p < 2.2 \times 10^{-16}$). VIFs range from 1.10 to 6.05; `Bene_Avg_Risk_Scre` (6.05) and `Bene_CC_PH_HF_NonIHD_V2_Pct` (5.60) are elevated but below the conventional threshold of 10. Diagnostic plots show the same pattern of violations as Model 1; the same asymptotic defense applies at $n = 3{,}752$.

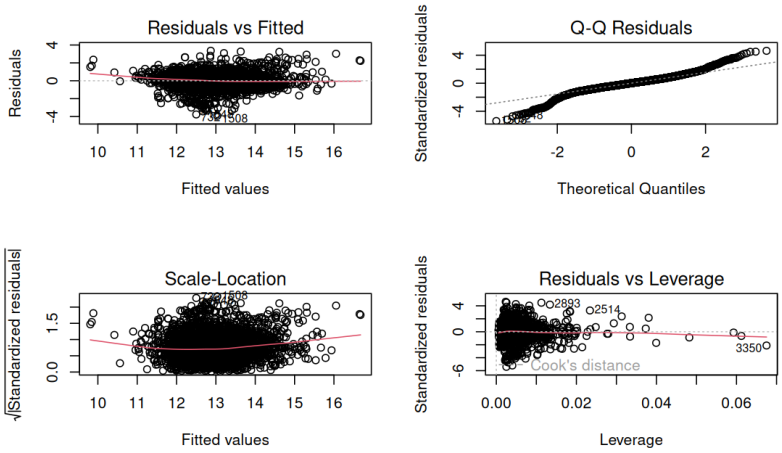

Figure 5: Model 2 diagnostic plots. Assumption violations mirror Model 1: mild non-linearity at high fitted values, heavy-tailed residuals, and heteroscedasticity.

Quantile Regression Extension, Model 2

The Model 2 quantile regression uses the same 12 predictors and kernel standard errors; the CC-complete subset's already-elevated median ($404,539) means these quantiles characterise heterogeneity *within* a high-volume stratum rather than across the full provider population.

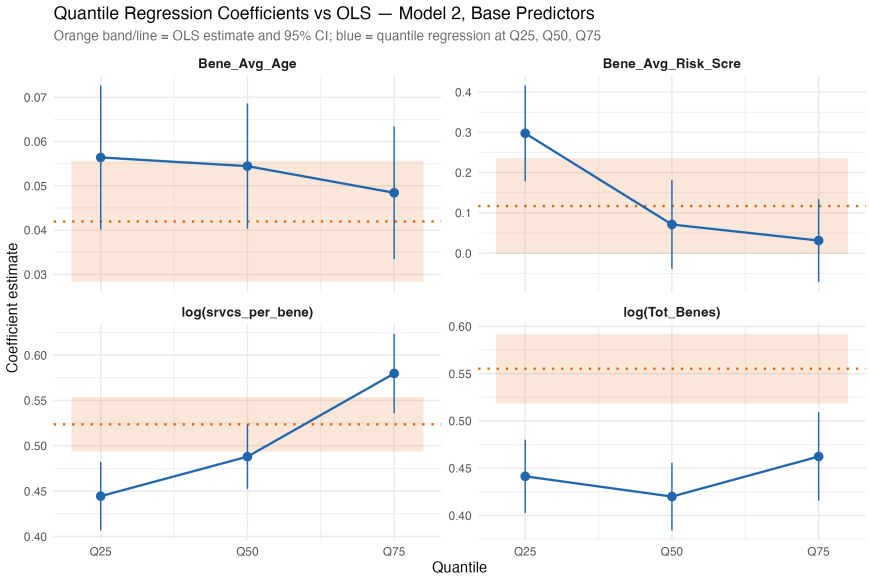

Figure 6: Quantile regression coefficients at Q25, Q50, and Q75 versus OLS for the four base predictors in Model 2 (CC-complete subset, $n = 3{,}752$). Orange band and dotted line represent the OLS estimate and 95% confidence interval. Blue points and lines show quantile regression estimates with 95% confidence intervals.

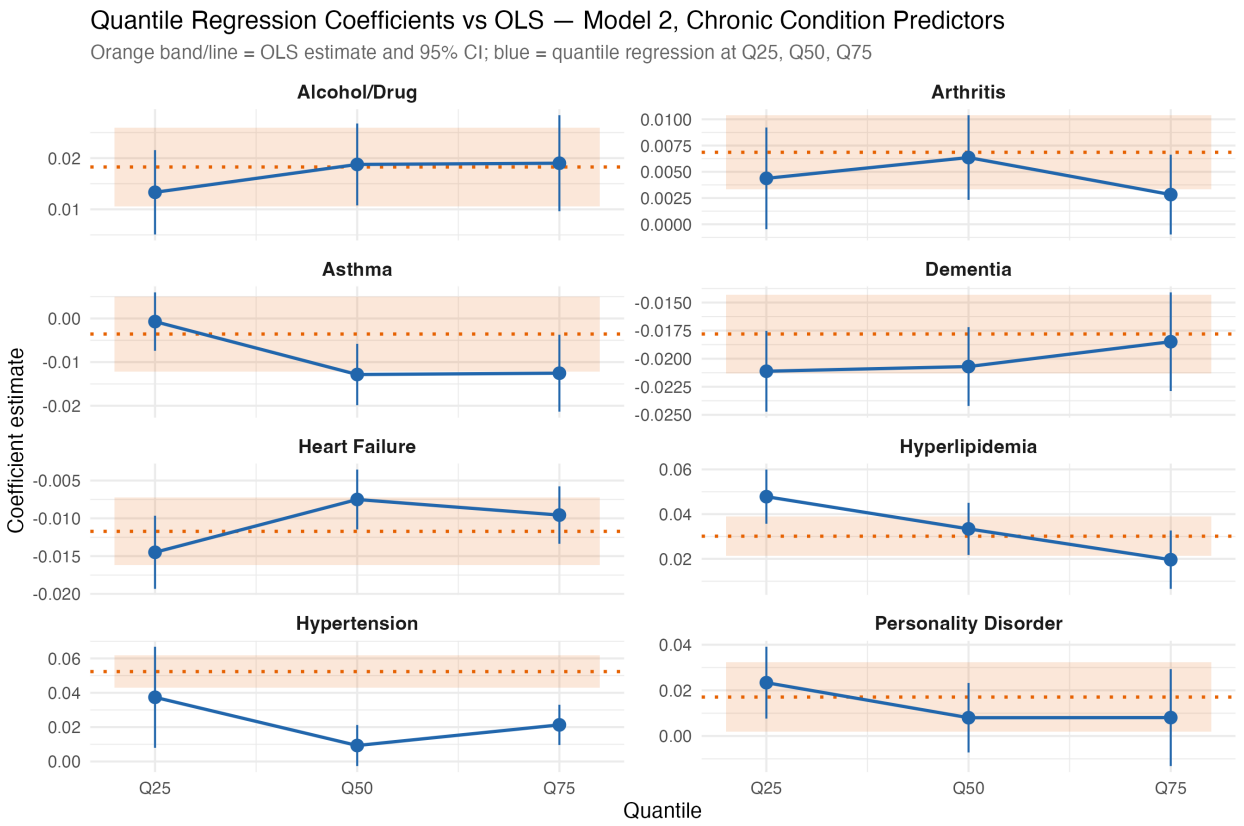

Figure 7: Quantile regression coefficients at Q25, Q50, and Q75 versus OLS for the eight chronic condition predictors in Model 2 (CC-complete subset, $n = 3{,}752$). Orange band and dotted line represent the OLS estimate and 95% confidence interval.

The base predictor results differ from Model 1 in two important respects. `log(Tot_Benes)` is flat across quantiles (0.441, 0.420, 0.462) but lies *below* the OLS estimate of 0.555 at all three, the reverse of Model 1. Volume has already done much of its work selecting providers into this high-gap stratum; within-stratum variation carries less marginal explanatory power than OLS suggests. `log(srvcs_per_bene)` reverses direction entirely: it increases from 0.444 at Q25 to 0.580 at Q75, with Q75 outside the OLS band of 0.524. Among high-volume providers, billing intensity is a distinguishing characteristic of over-billers specifically, not a uniform driver, the opposite of what Model 1 showed across the full population. `Bene_Avg_Risk_Scre` declines sharply from 0.297 at Q25 to 0.032 at Q75 (non-significant, $p = 0.548$): among over-billers, patient complexity adds almost nothing once condition mix is controlled for. `Bene_Avg_Age` is stable within the OLS band throughout.

Among the chronic condition predictors, three findings stand out. **Dementia** is the most consistent: negative across all quantiles (-0.021, -0.021, -0.018), all within the OLS band and all significant, confirming a uniform gap-reducing effect regardless of where a provider sits in the distribution. **Hypertension** shows an unexpected V-shape, 0.037 at Q25, near zero at Q50 (0.009, $p = 0.127$), recovering to 0.021 at Q75, revealing that the OLS headline result of 5.4% per percentage point applies to the extremes of the within-stratum distribution but not to typical high-volume providers. **Hyperlipidemia** declines from 0.048 at Q25 to 0.020 at Q75, its positive effect concentrated among under-billers. Alcohol/Drug, Heart Failure, Asthma, Personality Disorder, and Arthritis are broadly consistent with their OLS estimates throughout.

Taken together, the Model 2 quantile regression shows that billing intensity distinguishes over-billers from under-billers within the high-volume stratum in a way OLS misses, and that the headline hypertension finding is concentrated at the distributional extremes rather than typical providers. These results specify *for whom* within the CC-complete subset each predictor matters most.

Answer to Research Question

Provider volume is the dominant predictor across the full population ($R^2 = 0.629$); chronic condition mix adds meaningful explanatory power within the high-volume subset, with cardiovascular-metabolic conditions amplifying the gap and neurological conditions reducing it. Quantile regression reveals these OLS averages mask meaningful distributional heterogeneity, a pattern that recurs across all four analyses.

3.2 One-Way ANOVA

Research Question: Do average standardized Medicare payment gaps differ significantly across provider specialties in Massachusetts, and which specific specialties drive the largest gaps?

Exploratory Analysis

The ANOVA is restricted to the top 10 specialties by provider count ($n = 22{,}753$) to balance statistical reliability with interpretability. With 97 specialty groups in the full dataset, including all specialties would produce 4,656 pairwise comparisons under Games–Howell correction, rendering the post-hoc results uninterpretable. Restricting to the top 10 ensures each group has at least 1,033 providers while capturing the specialties that together account for the majority of Massachusetts Medicare billing activity.

Table 2 summarizes the payment gap distribution by specialty. The gap between extremes is substantial: Diagnostic Radiology has a median gap of $358,812. That is more than 15 times the $23,460 median for Licensed Clinical Social Workers. Provider count and payment gap rank diverge sharply: Nurse Practitioners are the most numerous specialty (5,496 providers) but rank eighth by median gap, while Diagnostic Radiology ranks eighth by count but first by gap.

Table 2: Payment gap summary statistics for the top 10 Massachusetts provider specialties by count, ordered by median payment gap.

Specialty	n	Median ($)	Mean ($)	P25 ($)	P75 ($)
Diagnostic Radiology	1,118	358,812	488,361	187,190	572,391
Anesthesiology	1,375	265,821	334,470	157,180	410,759
Emergency Medicine	1,299	169,158	223,101	77,657	296,636
Internal Medicine	3,978	118,226	168,631	47,865	218,482
Family Practice	1,335	86,743	131,131	18,157	181,964
Physical Therapist (Private)	2,005	64,477	88,434	33,993	115,239
Physician Assistant	3,070	57,653	109,971	20,910	123,101
Nurse Practitioner	5,496	49,358	86,151	16,198	111,916
Psychiatry	1,033	37,094	62,622	14,824	80,237
Licensed Clinical Social Worker	1,044	23,460	32,280	14,450	39,268

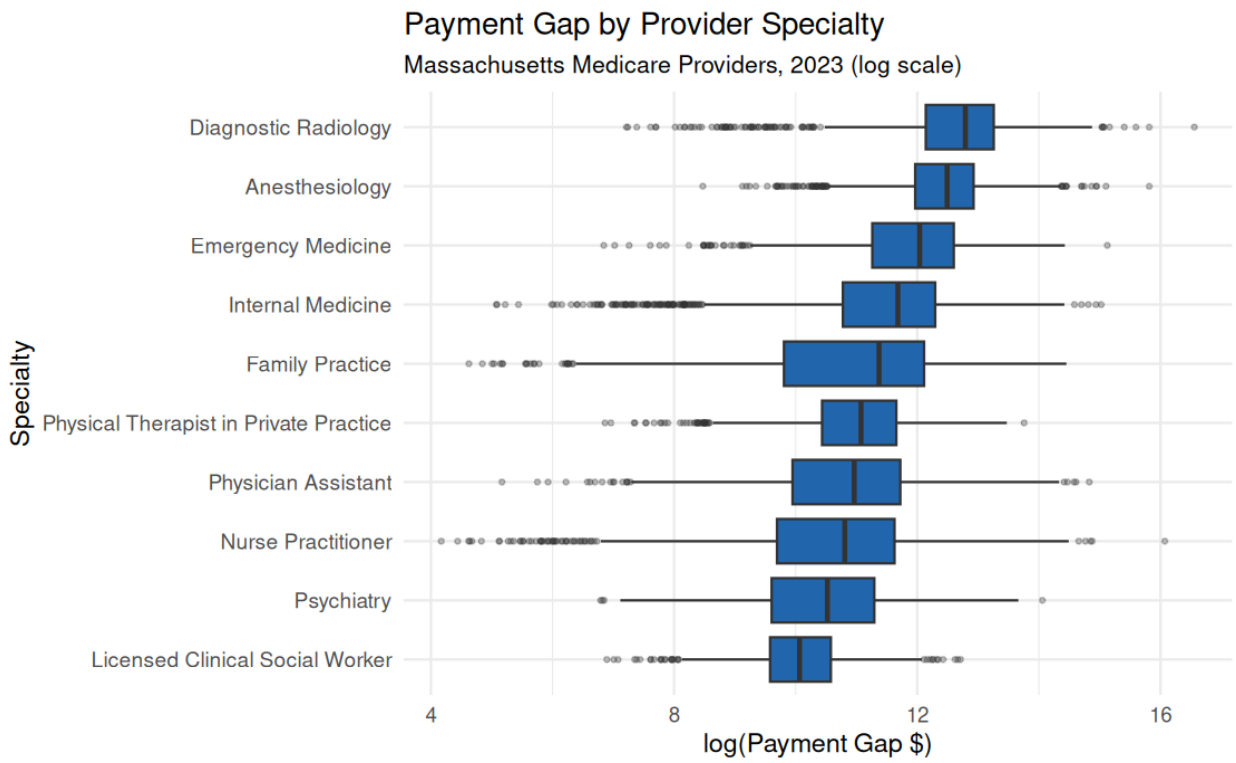

Figure 8: Distribution of log payment gap by provider specialty among the top 10 Massachusetts specialties (2023), ordered by median log payment gap.

Assumption Checks

Shapiro-Wilk rejects normality in all ten groups, though group sizes (1,033 to 5,496) render the F-test asymptotically valid. Levene's test rejects equal variances ($F(9, 21{,}743) = 110.08$, $p < 2.2 \times 10^{-16}$), so Welch's ANOVA and Games-Howell post-hoc tests are used throughout.

ANOVA Results

Welch's one-way ANOVA returns $F(9, 6{,}181.8) = 929.99$, $p < 2.2 \times 10^{-16}$, decisively rejecting the null hypothesis of equal mean log payment gaps across all ten specialties. The effect size is large: $\eta^2 = 0.2097$, meaning specialty group membership alone explains 21% of variance in log payment gap.

Table 3: Mean log payment gap and compact letter display by specialty. Groups sharing a letter are not significantly different at $\alpha = 0.05$ after Games–Howell correction.

Specialty	Mean log gap	Group
Diagnostic Radiology	12.541	b
Anesthesiology	12.384	a
Emergency Medicine	11.867	c
Internal Medicine	11.434	f
Physical Therapist (Private)	10.996	d
Family Practice	10.893	de
Physician Assistant	10.827	e
Nurse Practitioner	10.590	h
Psychiatry	10.418	i
Licensed Clinical Social Worker	10.041	g

Post-Hoc Tests

Games–Howell tests across all 45 pairwise combinations find 43 of 45 pairs significantly different at $\alpha = 0.05$. The two non-significant pairs are Family Practice vs. Physical Therapist ($p = 0.583$) and Family Practice vs. Physician Assistant ($p = 0.958$), forming a statistically indistinguishable middle tier. The largest contrast is Diagnostic Radiology vs. Licensed Clinical Social Worker ($\Delta = -2.500$ log units, $e^{2.500} \approx 12.2\times$ in raw dollars).

Non-Parametric Robustness Check: Kruskal–Wallis and Dunn's Test

To confirm that the parametric findings reflect genuine location differences rather than mean differences driven by extreme values, the analysis is complemented with a Kruskal–Wallis test and Dunn's post-hoc test with Bonferroni correction.

The Kruskal–Wallis test decisively rejects the null hypothesis that payment gap distributions are equal across all ten specialties ($\chi^2(9) = 5{,}285$, $p < 2.2 \times 10^{-16}$), in complete agreement with Welch's ANOVA ($F(9, 6{,}181.8) = 929.99$, $p < 2.2 \times 10^{-16}$).

Dunn's post-hoc tests with Bonferroni correction find 42 of 45 pairs significantly different, compared to 43 of 45 under Games–Howell. The two approaches agree on the core finding:

Family Practice vs. Physical Therapist remains non-significant under both methods. Two additional pairs are non-significant under Dunn's test but significant under Games–Howell: Physical Therapist vs. Physician Assistant, and Anesthesiology vs. Diagnostic Radiology.

The Physical Therapist vs. Physician Assistant discrepancy is unsurprising given their similar medians ($64,477 vs. $57,653); the rank-based test applies a more conservative criterion than the mean-based one. The Anesthesiology vs. Diagnostic Radiology discrepancy is more substantive: Games–Howell detects a difference in means ($266k vs. $359k) that Dunn's test does not confirm at the distributional level, suggesting the mean difference between these two high-gap specialties is partly driven by the extreme right tail of Diagnostic Radiology rather than a systematic shift in the bulk of the distribution.

Overall the two approaches converge on the same structural picture: a clear high-gap tier (Diagnostic Radiology, Anesthesiology, Emergency Medicine), a low-gap tier (Licensed Clinical Social Worker, Psychiatry), and a middle cluster whose internal boundaries are sensitive to whether mean-based or rank-based comparisons are used. The core specialty ranking is robust; the precise boundaries within the middle tier are not.

Full Specialty Landscape: All 97 Provider Types

The formal ANOVA is restricted to the top 10 specialties by provider count, raising the question of whether the 87 excluded specialties exhibit more extreme payment gaps than those analyzed. To address this, the median payment gap is computed for all 97 Massachusetts provider specialties and ranked from highest to lowest.

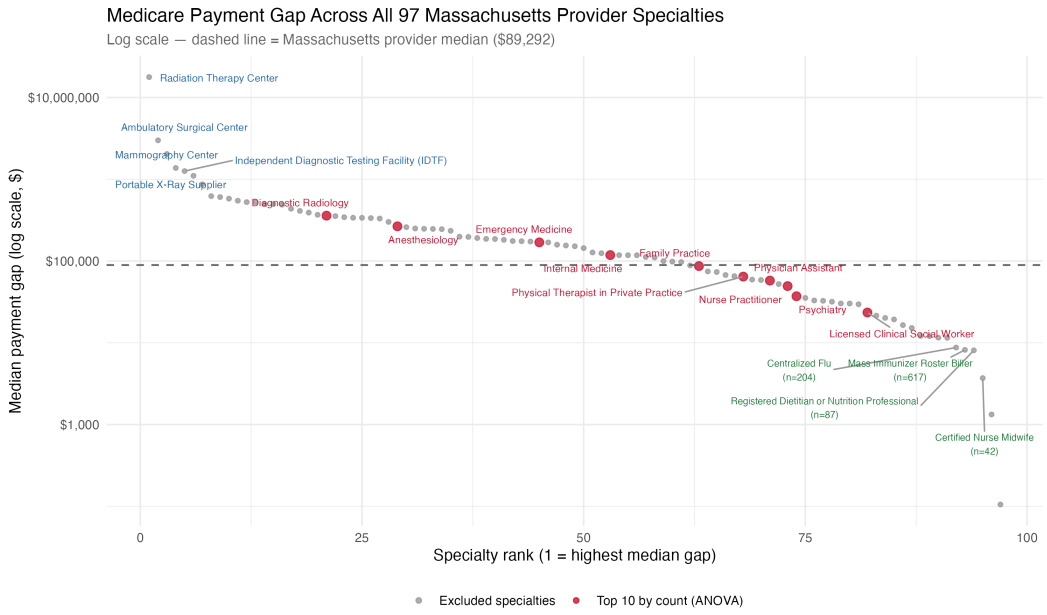

Figure 9: Median payment gap for all 97 Massachusetts provider specialties on a log scale, ranked from highest to lowest. Red points indicate the top 10 specialties by provider count included in the formal ANOVA. Blue labels identify facility and equipment-based outliers at the top of the distribution. Green labels identify specialties with near-Medicare billing at the bottom. The dashed line marks the Massachusetts provider median ($89,292).

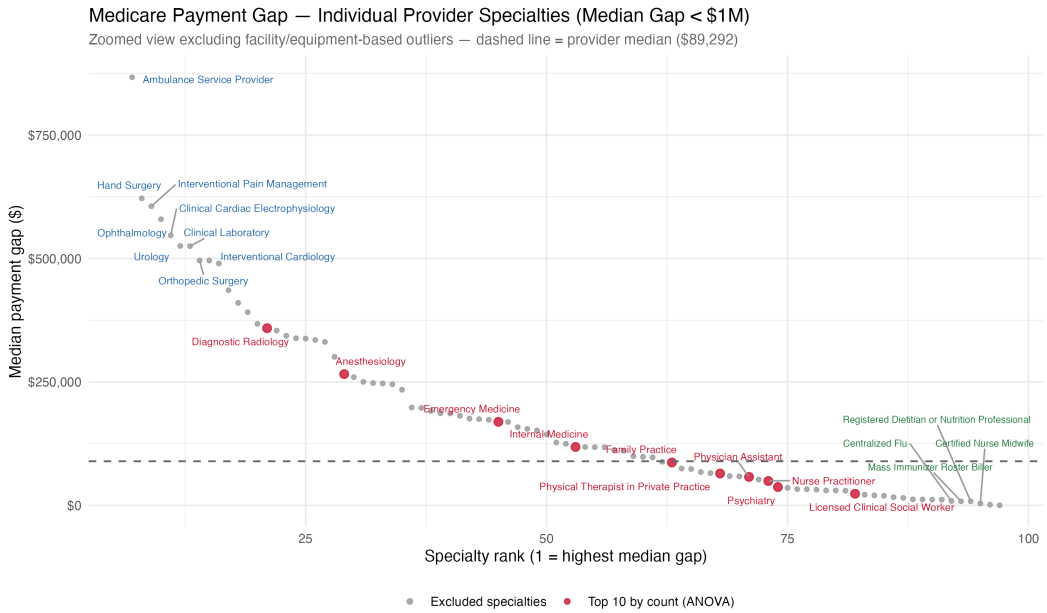

Figure 10: Median payment gap for Massachusetts provider specialties with median gaps below $1,000,000, excluding facility and equipment-based outliers. Blue labels identify excluded specialties ranking above Diagnostic Radiology ($358,812). Green labels identify specialties billing at or near Medicare standardized rates.

The five highest-gap entries are facility and equipment-based entities: Radiation Therapy Center ($17.8M), Ambulatory Surgical Center ($3.0M), Mammography Center ($2.0M), Portable X-Ray Supplier ($1.4M), and Independent Diagnostic Testing Facility ($1.3M), with median gaps one to two orders of magnitude larger than any individual physician specialty. Their exclusion from the ANOVA is appropriate on substantive as well as statistical grounds: these are institutional billers operating under a different billing logic than individual practitioners.

The zoomed plot reveals that at least nine excluded individual specialties have median gaps exceeding Diagnostic Radiology ($358,812), including Ambulance Service Provider ($867k), Hand Surgery ($622k), and Ophthalmology ($579k), all procedural or equipment-intensive, consistent with the ANOVA finding that procedure-intensive practice patterns drive the largest gaps. Diagnostic Radiology anchors the top among high-count specialties but ranks approximately 20th in the full landscape.

At the upper extreme, two distinct mechanisms produce large gaps: procedure diversity among individual specialists (Hand Surgery and Vascular Surgery each bill a median of 83.5 codes) and per-procedure unit cost among facility-based entities (Ambulance Service Providers bill a median of 4 codes yet achieve a $867k gap; Mammography Centers bill 7 codes with a $2.0M gap). Oversight strategies targeting individual practitioners can reasonably use procedure diversity as a screening signal, but facility and equipment-based billers require a different analytical lens.

At the bottom, specialties with narrow standardized billing profiles anchor near-Medicare gaps: Certified Nurse Midwives (median $3,704, 8 codes) and Mass Immunizer Roster Billers ($8,194, 12 codes) leave limited room to inflate submitted charges, consistent with the RQ3 finding that practice breadth (`log(Tot_HCPCS_Cds)`, OR = 3.77) is among the strongest predictors of high-gap classification.

Answer to Research Question

Specialty group membership alone explains 21% of variance in log payment gap, with Diagnostic Radiology, Anesthesiology, and Emergency Medicine anchoring the high end and Licensed Clinical Social Workers and Psychiatry the low end; two structurally distinct mechanisms, procedure diversity and per-procedure unit cost, drive the largest gaps.

3.3 Logistic Regression

Research Question: Do provider structure, billing behavior, and low-income patient burden predict whether a Massachusetts provider falls in the high payment gap group, and which factors have the strongest independent effect on the odds of being a high-gap provider?

The binary outcome `high_gap` is coded 1 for providers above the Massachusetts median payment gap and 0 otherwise, yielding a balanced split of 17,737 high-gap and 10,819 low-gap providers in the RQ3 analysis dataset ($n = 28{,}556$, after excluding providers with suppressed `Bene_Dual_Cnt`).

Predictor Selection

Six predictors spanning billing behavior (log(srvcs_per_bene), drug provider), patient complexity (log(Tot_Benes), Bene_Avg_Risk_Scre), and practice structure (log(Tot_HCPCS_Cds), pct_dual) are included. Rndrng_Prvdr_Ent_Cd was excluded: 95.7% of providers are individual practitioners, leaving too few organizational providers for reliable estimation.

Exploratory Analysis

Table 4 compares the median value of each continuous predictor between low-gap and high-gap providers. The largest separation is observed for `Tot_HCPCS_Cds`: high-gap providers bill a median of 29 distinct procedure codes compared to 11 for low-gap providers, a 2.6-fold difference suggesting that practice breadth is strongly associated with high-gap classification. Provider volume (`Tot_Benes`) also differs substantially: high-gap providers serve a median of

315 beneficiaries versus 99 for low-gap providers. Patient complexity (`Bene_Avg_Risk_Scre`) is modestly higher among high-gap providers (1.476 vs. 1.298), consistent with the MLR finding that sicker patient panels generate larger gaps. Billing intensity (`srvcs_per_bene`) shows a smaller separation (2.56 vs. 2.07). Notably, dual eligibility proportion (`pct_dual`) is *lower* among high-gap providers (0.224 vs. 0.310), suggesting that providers serving more low-income panels are less likely to be high-gap classifiers.

Table 4: Median predictor values by `high_gap` classification in the RQ3 analysis dataset ($n = 28{,}556$).

Predictor	Low-gap ($n = 10{,}819$)	High-gap ($n = 17{,}737$)
`Tot_Benes`	99	315
`srvcs_per_bene`	2.07	2.56
`Bene_Avg_Risk_Scre`	1.298	1.476
`pct_dual`	0.310	0.224
`Tot_HCPCS_Cds`	11	29

The binary predictor `drug_provider` also shows a meaningful separation: 72.0% of providers who bill drug charges are high-gap classifiers, compared to 59.6% of non-drug providers, a 12.4 percentage point difference suggesting that drug billing is associated with higher odds of exceeding the median payment gap.

Model Results

All six predictors are statistically significant at $p < 2 \times 10^{-16}$. Odds ratios with 95% confidence intervals are reported in Table 5.

Table 5: Logistic regression odds ratios with 95% confidence intervals. All predictors significant at $p < 2 \times 10^{-16}$.

Predictor	OR	95% CI Low	95% CI High
`log(Tot_Benes)`	6.7236	6.3579	7.1158
`log(srvcs_per_bene)`	2.6517	2.5267	2.7841
`Bene_Avg_Risk_Scre`	2.2853	2.1632	2.4156
`pct_dual`	0.2120	0.1706	0.2631
`drug_provider`	0.1774	0.1598	0.1969
`log(Tot_HCPCS_Cds)`	3.7665	3.5718	3.9739

Coefficients are interpreted on the odds ratio scale; holding all other variables constant:

- **Provider volume** (`log(Tot_Benes)`): the strongest predictor by z-value ($z = 66.34$); each doubling of beneficiaries served multiplies the odds of high-gap classification by 3.75 (OR = 6.72, 95% CI: [6.36, 7.12]).
- **Practice breadth** (`log(Tot_HCPCS_Cds)`): each doubling of distinct procedure codes billed is associated with a 2.51-fold increase in odds (OR = 3.77, 95% CI: [3.57, 3.97]).
- **Billing intensity** (`log(srvcs_per_bene)`): each doubling of services per beneficiary is associated with a 1.97-fold increase in odds (OR = 2.65, 95% CI: [2.53, 2.78]).
- **Patient complexity** (`Bene_Avg_Risk_Scre`): a one-unit increase in HCC risk score is associated with a 2.29-fold increase in odds (OR = 2.29, 95% CI: [2.16, 2.42]).
- **Low-income patient burden** (`pct_dual`): a one-unit increase in dual-eligibility proportion is associated with a 78.8% decrease in odds of high-gap classification (OR = 0.212, 95% CI: [0.171, 0.263]).
- **Drug billing** (`drug_provider`): providers billing drug charges have 82.3% lower odds than non-drug providers (OR = 0.177, 95% CI: [0.160, 0.197]). This reverses the positive

univariate association (72.0% vs. 59.6% high-gap rate), indicating drug billing proxies for characteristics that reduce gap odds once other dimensions are controlled for.

McFadden pseudo-$R^2 = 0.4391$, indicating strong model fit by conventional standards (values above 0.2 are considered excellent for logistic regression). The model AIC is 21,267.

A note on model convergence: A convergence warning (`fitted probabilities numerically 0 or 1 occurred`) reflects the strength of the signal at volume and practice breadth extremes rather than a model pathology; estimates and standard errors are reliable.

Confusion Matrix & Performance Metrics

The optimal classification threshold was selected using Youden's J statistic (maximizing sensitivity + specificity -1), yielding a threshold of 0.609. The resulting confusion matrix is shown in Table 6.

Table 6: Confusion matrix at threshold = 0.609.

	Predicted: 0	Predicted: 1
Actual: 0	TN = 8,814	FP = 2,005
Actual: 1	FN = 2,980	TP = 14,757

The threshold of 0.609 lies slightly above 0.5, reflecting the mild class imbalance in the dataset (62% high-gap providers). At this threshold the model achieves accuracy of 82.5%, sensitivity of 83.2%, specificity of 81.5%, and precision of 88.0%. Sensitivity and specificity are nearly equal, indicating the model is well calibrated for balanced detection of both classes under Youden's criterion.

ROC Curve & AUC

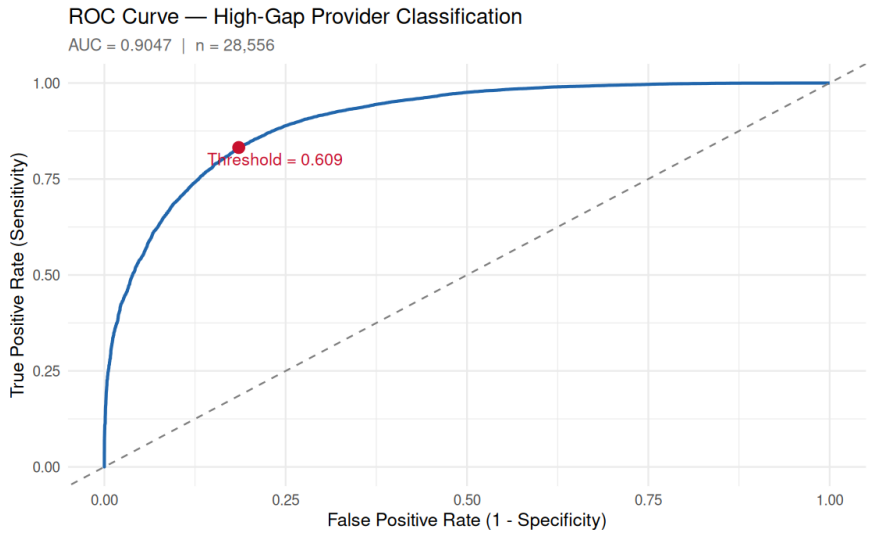

Figure 11: ROC curve for high-gap provider classification. The dashed diagonal represents a random classifier (AUC = 0.5). The filled point marks the optimal threshold of 0.609 selected by Youden's J statistic. AUC = 0.9047. Model: `high_gap` ∼ `log(Tot_Benes)` + `log(srvcs_per_bene)` + `Bene_Avg_Risk_Scre` + `pct_dual` + `drug_provider` + `log(Tot_HCPCS_Cds)`.

The model achieves AUC = 0.9047, if a high-gap and a low-gap provider are drawn at random, the model assigns the higher probability to the correct provider 90.5% of the time. The ROC curve rises steeply toward the upper-left corner before flattening, confirming strong discrimination at low false positive rates.

Sensitivity Analysis: Classification Threshold Robustness

The `high_gap` outcome is defined at the Massachusetts provider median, an administrative convenience rather than a clinically or economically meaningful cutoff.

To assess whether the logistic regression findings are sensitive to this choice, the model is refit at four alternative thresholds: the 40th, 50th, 60th, and 75th percentiles of the payment gap distribution in the RQ3 analysis dataset ($95,769, $132,121, $178,079, and $292,776 respectively). All six predictors and the full model specification are held constant across

refits; only the outcome definition changes.

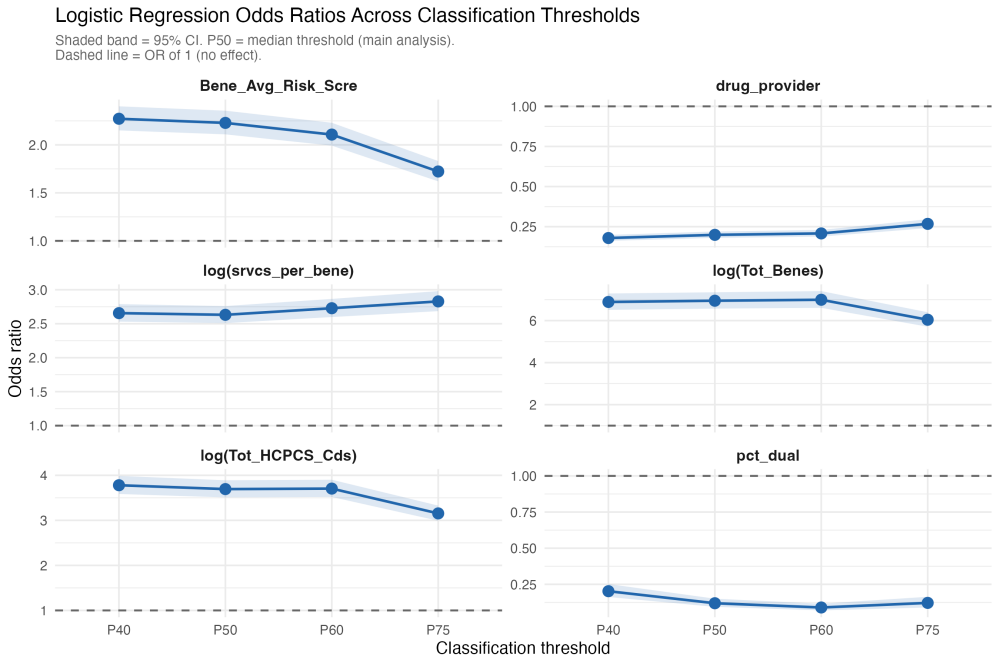

Figure 12: Logistic regression odds ratios at four classification thresholds (P40, P50, P60, P75). P50 corresponds to the main analysis. Shaded bands show 95% confidence intervals. The dashed line marks OR = 1 (no effect). Each panel shows one predictor; the y-axis is free-scaled across panels.

The results confirm that the logistic regression findings are robust to the choice of classification threshold. Every predictor maintains the same sign and approximate magnitude across all four thresholds, and no finding approaches the null (OR = 1) at any threshold. Several patterns are worth noting.

`log(Tot_Benes)` is the most stable predictor, with odds ratios of 6.89, 6.95, and 6.99 at P40 through P60, attenuating modestly to 6.04 at P75. Provider volume dominates high-gap classification at every threshold but is slightly less decisive at the very top of the distribution, consistent with the quantile regression finding that volume is a near-uniform driver across the conditional gap distribution.

`pct_dual` shows the most substantively interesting pattern across thresholds. The negative association with high-gap classification becomes more pronounced as the threshold rises:

OR = 0.202 at P40, 0.119 at P50, 0.090 at P60, then recovering slightly to 0.121 at P75. Providers serving panels with higher proportions of dual-eligible patients are increasingly unlikely to be classified as high-gap as the definition tightens, with the association strongest at P60 rather than at the extremes. The OR remains far below 1 at every threshold and confidence bands never approach the null, confirming the dual-eligibility finding is not an artifact of the median split.

`log(srvcs_per_bene)` shows a mild upward trend from 2.66 at P40 to 2.83 at P75, suggesting billing intensity is slightly more discriminating at higher thresholds, consistent with the Model 1 quantile regression finding that billing intensity matters more for providers at the upper tail of the gap distribution.

`Bene_Avg_Risk_Scre`, `log(Tot_HCPCS_Cds)`, and `drug_provider` all attenuate somewhat at P75 but remain strongly significant throughout. The attenuation at P75 is consistent across these three predictors and suggests that the top quarter of the gap distribution is somewhat less differentiated by patient complexity, practice breadth, and drug billing status than the broader high-gap group. Volume and billing intensity become relatively more dominant at the extreme top.

Specialty Robustness Check

As a further robustness check, the logistic regression was refit adding specialty group membership as a covariate (top 10 specialties by provider count plus an *Other* category). All six core findings are stable or strengthen after specialty adjustment. Notably, the `pct_dual` odds ratio moves from 0.212 to 0.168, confirming that the dual-eligibility association is not explained by the tendency of high dual-eligibility providers to cluster in inherently low-gap specialties. `log(Tot_Benes)` strengthens from 6.72 to 7.73, indicating that within a given specialty, volume is an even stronger differentiator of billing behavior. `log(Tot_HCPCS_Cds)` attenuates modestly from 3.77 to 3.62, consistent with practice breadth and specialty being partially correlated. McFadden pseudo-R^2 improves from 0.439 to 0.462 and AUC from 0.905

to 0.913.

Answer to Research Question

All six predictors significantly predict high-gap classification (AUC = 0.905, 82.5% accuracy). Provider volume dominates (OR = 6.72), followed by practice breadth (OR = 3.77) and billing intensity (OR = 2.65); dual-eligibility burden is the strongest inverse predictor (OR = 0.21), strengthening to OR = 0.168 after specialty adjustment and remaining far from the null across all four threshold refits.

Additional Analysis: Bootstrap & Permutation Tests

Research Question: Does the median standardized Medicare payment gap differ significantly between Massachusetts providers whose patient panels have low versus high proportions of dual-eligible beneficiaries?

Exploratory Analysis

Providers are split at the median `pct_dual` into two equal groups of 14,278 providers each. Providers above the median dual-eligibility proportion are classified as *high dual-eligibility*; those at or below the median are classified as *low dual-eligibility*.

Table 7 reports the payment gap distribution for each group. Low dual-eligibility providers have a median gap of $176,569 compared to $99,942 for high dual-eligibility providers, a difference of $76,627, or 76.7% larger among providers serving fewer dual-eligible patients.

Table 7: Payment gap summary statistics by dual-eligibility group (n = 14,278 per group).

Group	n	Median ($)	Mean ($)	P25 ($)	P75 ($)
Low dual-eligibility	14,278	176,569	386,708	72,552	381,801
High dual-eligibility	14,278	99,942	235,487	43,608	213,918

The mean gap is substantially larger than the median in both groups, confirming the heavy right skew documented in the preprocessing analysis. The median is therefore the appropriate

measure of central tendency for this comparison.

Continuous Analysis: Validating the Median Split

The binary split at the median `pct_dual` is an analytical convenience. To assess whether this choice masked a threshold effect or non-linearity in the underlying relationship, `pct_dual` is examined as a continuous predictor of `log(payment_gap)` using univariate quantile regression at Q25, Q50, and Q75 alongside an OLS reference line.

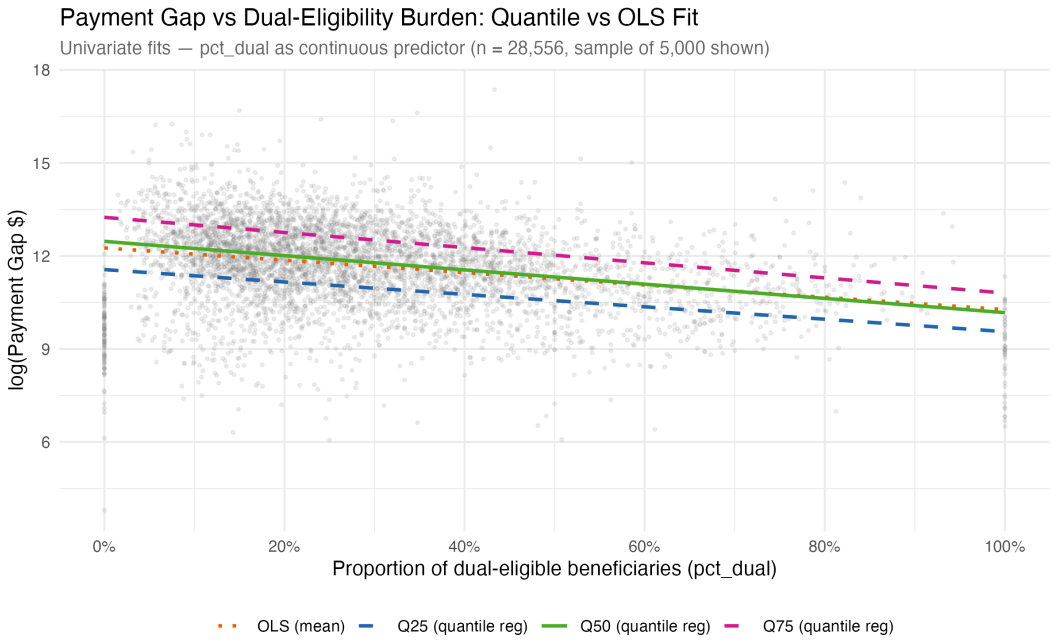

Figure 13: Univariate quantile regression fits at Q25, Q50, and Q75 for `log(payment_gap)` on `pct_dual` as a continuous predictor ($n = 28{,}556$; sample of 5,000 shown). OLS reference line shown as orange dotted. Straight parallel lines confirm a linear, uniform relationship across the full range of dual-eligibility burden.

All four lines are straight and decline consistently across the full range of `pct_dual` from 0% to 100%. There is no visible curvature, threshold effect, or acceleration at any point. The quantile lines are parallel throughout, indicating that dual-eligibility burden shifts the entire payment gap distribution downward uniformly, affecting low-gap and high-gap providers to the same degree. OLS and Q50 are nearly identical, confirming the mean and median tell the same story at each level of dual-eligibility burden.

Analysis & Results

Bootstrap Confidence Intervals Bootstrap confidence intervals for the median payment gap were constructed separately for each group using 10,000 replicates. Results are summarized in Table 8 and visualized in Figure 14.

Table 8: Bootstrap 95% confidence intervals for the median payment gap by dual-eligibility group ($B = 10{,}000$ replicates).

Group	Observed Median ($)	CI Lower ($)	CI Upper ($)
Low dual-eligibility	176,569	172,148	181,248
High dual-eligibility	99,942	97,686	102,517

The two confidence intervals do not overlap: the upper bound for the high dual-eligibility group ($102,517) falls nearly $70,000 below the lower bound for the low dual-eligibility group ($172,148), confirming the difference is not an artifact of sampling variability.

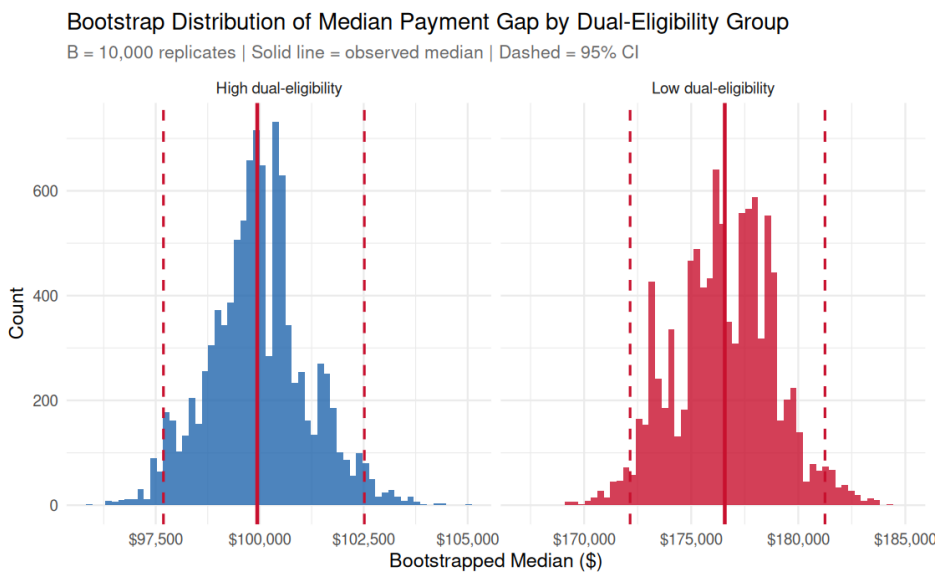

Figure 14: Bootstrap distributions of the median payment gap for low (red) and high (blue) dual-eligibility providers ($B = 10{,}000$ replicates each). Solid vertical lines mark the observed medians; dashed lines mark the 95% confidence interval bounds. The complete separation between the two distributions confirms that the difference in medians is statistically reliable.

Permutation Test Under the null hypothesis that dual-eligibility group membership is unrelated to the payment gap, group labels were permuted 10,000 times and the difference

in medians recomputed on each permutation. The observed difference of \$76,628 was not exceeded by any of the 10,000 permuted differences, yielding a permutation p-value of $p < 0.0001$. The null hypothesis is rejected.

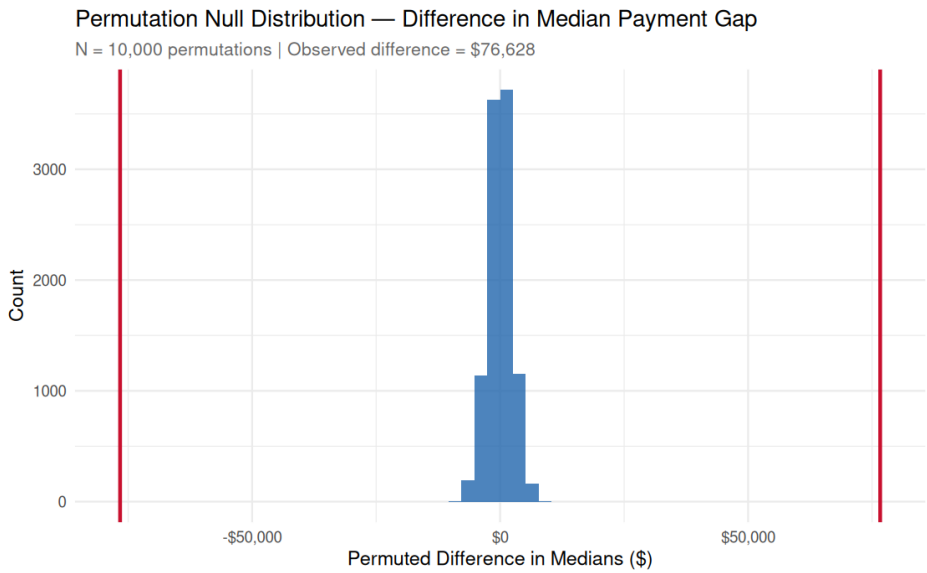

Figure 15: Empirical null distribution of 10,000 permuted differences in median payment gap. The null distribution is concentrated tightly around zero; the red vertical lines marking the observed difference of \$76,628 and its negative mirror fall entirely outside the range of the null distribution. No permuted difference approached the observed value, yielding $p < 0.0001$.

Comparison to Parametric Alternative The Wilcoxon rank-sum test with continuity correction returns $W = 77{,}130{,}046$, $p < 2.2 \times 10^{-16}$, confirming the permutation test result. The Hodges-Lehmann estimate of the location shift is $-\$57{,}980$ (95% CI: $[-\$61{,}574,\ -\$54{,}436]$), indicating that a randomly selected high dual-eligibility provider has a median payment gap approximately \$57,980 lower than a randomly selected low dual-eligibility provider. This is directionally consistent with the bootstrap and permutation findings, though smaller in magnitude than the raw difference in group medians (\$76,628). Both tests reach the same conclusion.

Answer to Research Question

The median payment gap is \$76,628 lower among high dual-eligibility providers (\$99,942 vs. \$176,569); bootstrap CIs do not overlap and the permutation p-value is < 0.0001. This unconditional finding complements the logistic regression result: providers serving more dual-eligible patients have both lower odds of high-gap classification and structurally lower median gaps, a pattern that holds both conditionally and unconditionally.

4 Discussion

A Coherent Story Across Four Analyses

Taken together, the four analyses converge on a coherent account of what drives the Medicare payment gap among Massachusetts providers, and what does not.

The payment gap is primarily a function of what providers do and who their patients are, not of geography or random variation. Provider volume is the dominant predictor across the full provider population (RQ1, $R^2 = 0.629$). Specialty group membership explains an additional 21% of variance independently (RQ2). Volume is also the strongest predictor of high-gap classification by z-value (RQ3), followed by practice breadth and billing intensity. These three dimensions paint a consistent picture: the payment gap is largest among high-volume procedural specialists billing across a wide range of services.

The Condition Mix Finding

A secondary but substantively important finding from RQ1 is that not all chronic conditions drive the gap in the same direction. Cardiovascular-metabolic conditions, hypertension and hyperlipidemia, are positively associated with the payment gap, while neurological and pulmonary conditions, dementia, heart failure, and asthma, are negatively associated, holding all other predictors constant. This asymmetry suggests that the composition of a provider's patient panel, not just its overall complexity, shapes billing behavior in ways the composite

HCC risk score cannot capture.

The hypertension finding is particularly striking. Each additional percentage point of hypertension prevalence is associated with a 5.4% larger payment gap on average, though quantile regression reveals this effect is concentrated at the extremes of the high-volume provider distribution and near zero for typical providers, qualifying the OLS result substantially. Hypertension is the most prevalent chronic condition in the Medicare population and one of the most manageable, primarily treated with inexpensive generics and routine monitoring. Yet, providers whose panels have higher hypertension prevalence submit systematically larger gaps, independent of volume, age, billing intensity, and all other retained chronic conditions. Whether this reflects comorbid procedures accompanying hypertension management, specialty selection effects, or something else cannot be determined from this data, but the signal warrants further investigation.

The Dual-Eligibility Finding and Its Implications

The most counterintuitive finding in this study emerges from RQ3 and RQ4. Providers serving patient panels with higher proportions of dual-eligible beneficiaries are substantially less likely to be high-gap classifiers (OR = 0.212) and have a median payment gap $76,628 lower than providers serving panels with fewer dual-eligible patients. This pattern holds both after controlling for practice breadth, patient complexity, billing intensity, and drug billing (RQ3), and unconditionally in a direct group comparison (RQ4).

The conventional expectation might be the opposite: providers serving the most economically vulnerable patients face higher administrative burden, thinner margins, and more billing complexity, conditions that might push toward higher submitted charges to compensate. The data does not support this expectation. Providers serving high dual-eligibility panels submit charges closer to Medicare's standardized payment rates, not further from them. This finding is robust: it survives refitting the logistic model at four different classification thresholds, strengthens rather than attenuates after controlling for specialty composition, and reflects a

linear and uniform continuous relationship across the full range of dual-eligibility burden.

One plausible mechanism is structural. Dual-eligible patients have their Medicare cost-sharing covered by Medicaid as secondary payer, reducing the gap between what Medicare pays and what the patient can contribute out of pocket. Providers serving these panels may face less billing uncertainty and therefore less incentive to set submitted charges aggressively. But this study cannot confirm that mechanism, it is offered as a hypothesis, not a conclusion.

What the data does establish is that the income profile of a provider's patient panel is a meaningful and statistically robust predictor of how aggressively that provider submits charges relative to Medicare's standardized rates. This observation raises a legitimate question: if providers serving the most comprehensively insured, lowest-income patients bill closest to CMS rates, what drives the substantially higher submitted charges among providers serving less vulnerable panels? The data cannot answer this question, but it is precisely the kind of question that Medicare billing oversight programs should be positioned to investigate.

Practical Implications

The findings have practical relevance for Medicare billing oversight and resource allocation.

The logistic regression model (RQ3) correctly classifies four in five providers as high-gap or low-gap using only six readily observable characteristics, provider volume, billing intensity, patient complexity, practice breadth, drug billing status, and dual-eligibility burden. All six are available in the standard CMS Physician and Other Practitioners dataset without any additional data linkage. This suggests that a simple screening model could identify providers warranting closer review with reasonable accuracy, without requiring claims-level audit data.

The specialty finding (RQ2) is equally actionable. Diagnostic Radiology, Anesthesiology, and Emergency Medicine account for a disproportionate share of the payment gap relative to their provider counts. The full specialty landscape analysis further suggests that oversight of facility and equipment-based entities requires a separate analytical lens, their gaps are

driven by per-procedure unit costs rather than the billing breadth and volume patterns that characterize individual practitioners. Conversely, high dual-eligibility providers in cognitive and behavioral health specialties provide a natural reference benchmark for providers billing closest to CMS rates.

Unexpected Results

Two findings require careful interpretation. Billing intensity's effect on the gap is confounded by condition mix and only emerges after chronic condition controls are added. Drug billing shows a positive univariate association but a negative conditional association (OR = 0.177), indicating it proxies for practice breadth and specialty, characteristics already controlled for in the model.

5 Limitations

This study draws on a large administrative dataset and a rigorous multi-method design, but several limitations qualify the conclusions drawn from each analysis.

5.1 Research Question 1, Multiple Linear Regression

Violation of LINE assumptions. Despite log transformation, MLR diagnostic plots reveal persistent non-linearity, heavy-tailed residuals, and heteroscedasticity reflecting the structural heterogeneity of the Massachusetts provider population. Quantile regression extensions address the heteroscedasticity concern directly; individual-level inference should be treated with caution.

Selection bias in Model 2. The complete-CC subset differs systematically from the full population (median gap $404,539 vs. $89,292; median risk score 1.727 vs. 1.326), so Model 2 answers a different question, what predicts the gap among high-volume providers, and its findings do not generalize to the broader Massachusetts provider population.

5.2 Research Question 2, One-Way ANOVA

Assumption violations. Levene's test decisively rejects equal variances ($F(9, 21{,}743) = 110.08$, $p < 2.2 \times 10^{-16}$) and Shapiro–Wilk rejects normality in all ten groups. Welch's ANOVA and Games–Howell post-hoc tests address the variance violation; a non-parametric robustness check using Kruskal–Wallis and Dunn's test confirms the specialty tier structure is not an artifact of distributional assumptions.

Restriction to top 10 specialties. Restricting to the top 10 specialties by count was a presentational necessity; a descriptive analysis of all 97 confirms the findings hold in the broader landscape.

5.3 Research Question 3, Logistic Regression

Arbitrary classification threshold. The `high_gap` indicator is defined at the Massachusetts provider median, an administrative convenience rather than a clinically meaningful cutoff. A sensitivity analysis at the 40th, 60th, and 75th percentile thresholds confirms all findings are robust to this choice.

Missing data exclusions. Complete-case analysis for `pct_dual` excludes 9,152 providers (∼25%) with suppressed `Bene_Dual_Cnt`. Because CMS suppresses `Bene_Dual_Cnt` when a provider has 10 or fewer dual-eligible beneficiaries, excluded providers are systematically small-panel practitioners, those serving the fewest and likely most vulnerable patients. The logistic regression findings therefore generalize to providers with at least moderate patient volumes and should not be extrapolated to solo or low-volume practitioners. Multiple imputation would be the principled remedy, but the suppression mechanism is structural rather than random: the missing values are not missing at random (MAR) but missing by design (censored below a threshold), which limits the validity of standard imputation approaches and is an acknowledged constraint of working with CMS public use data.

Independence assumption. Providers within the same hospital system or geographic market may have correlated billing patterns. Without a group practice identifier, standard errors cannot be clustered at the organizational level, and confidence intervals for all coefficients may be narrower than they should be.

5.4 Research Question 4, Bootstrap and Permutation Tests

Median split on `pct_dual`. Splitting providers at the median dual-eligibility proportion treats the comparison as binary. A continuous analysis using univariate quantile regression confirms the relationship is linear and uniform across the full range of `pct_dual`, validating the median split as a reasonable representation of a smooth continuous gradient.

Unconditional comparison conflates provider types. RQ4 compares median payment gaps without controlling for specialty, billing intensity, or patient complexity. The observed $76,628 difference therefore reflects not only the effect of dual-eligibility burden but also any systematic differences in provider type between the two groups. The logistic regression in RQ3 addresses this by conditioning on other predictors; RQ4 quantifies the raw unconditional gap.

Bootstrap confidence interval assumes i.i.d. resampling. If spatial autocorrelation or within-system clustering is present, the i.i.d. assumption is violated and the bootstrap confidence intervals may be too narrow. The reported intervals should be understood as lower bounds on uncertainty rather than fully valid distribution-free intervals.

6 Conclusions

This study set out to ask whether the Medicare payment gap among Massachusetts providers is structured or random, whether it follows predictable patterns tied to provider behavior and patient characteristics, or whether it is simply noise in a complex billing system. The answer is unambiguous: the gap is structured, measurable, and largely predictable from six variables

available in a single public dataset, and this structure holds up under multiple robustness checks including quantile regression, non-parametric testing, threshold sensitivity analysis, and specialty adjustment.

Three actionable recommendations follow from the findings.

First, Medicare billing oversight programs should prioritize procedurally intensive specialties, particularly Diagnostic Radiology, Anesthesiology, and Emergency Medicine among high-count individual providers, and facility and equipment-based entities such as Ambulatory Surgical Centers and Radiation Therapy Centers where per-procedure unit costs drive gaps an order of magnitude larger. A specialty-targeted review strategy would concentrate audit resources where the expected return is highest, but oversight of facility-based billers requires a different analytical lens than that designed for individual practitioners.

Second, the logistic model developed in RQ3 demonstrates that a simple six-variable screening tool can correctly classify four in five Massachusetts providers as high-gap or low-gap without any claims-level data. CMS and state-level oversight bodies could deploy a similar model as a low-cost triage layer before committing to full audits. Third, the dual-eligibility finding warrants direct policy attention: providers serving the most economically vulnerable patient panels bill closest to CMS standardized rates, while providers serving less vulnerable panels submit substantially higher charges. Understanding why this inversion exists, and whether it reflects billing incentives, structural differences in practice organization, or something else, is a question that Medicare payment reform efforts should be positioned to answer.

The broader implication is methodological. Administrative claims data, often treated as a byproduct of billing rather than a source of analytic signal, contains enough structure to support targeted, evidence-based oversight. The payment gap is not a mystery, it has predictors, it has a specialty profile, and it has a patient-panel signature. Acting on that structure is a policy choice, not a data limitation.

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Appendix A R Analysis Code

The complete analysis code and repository are available at:

https://github.com/danysigha/shape-of-the-break

All analyses were conducted in R. The preprocessing pipeline is available as a separate file (`01_preprocessing.Rmd`); the main analysis is in `02_analysis.Rmd`.